

ROSINA MARTHA DE HEER

BIOMEDICAL ENGINEER

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PROFESSIONAL SUMMARY

Analytical and detail-oriented Biomedical Engineer with a strong foundation in physiological systems modeling, biomedical instrumentation, and health data processing. Proven ability to apply engineering principles, mathematical modeling, and hardware-software integration to solve biological and clinical challenges. Dedicated to optimizing medical equipment, enhancing diagnostic accuracy, and driving innovation in non-invasive healthcare technology.

CORE COMPETENCIES & TECHNICAL SKILLS

Biomedical Instrumentation

Diagnostic equipment evaluation, medical sensor integration, and signal conditioning circuits.

Systems Modeling & Simulation

Physiological system dynamics, circuit simulation, and algorithmic data modeling.

Programming & Software

MATLAB, Python, C/C++, HTML/CSS, Microsoft Excel data processing pipelines.

Laboratory & Prototyping

Breadboard testing, microcontrollers (Arduino), Tinkercad, Proteus circuit design.

EDUCATION

Bachelor of Science (B.Sc.) in Biomedical Engineering

Graduation: [Year]

Kwame Nkrumah University of Science and Technology (KNUST) – Kumasi, Ghana

- Relevant Coursework:** Medical Instrumentation, Signals & Systems in Medicine, Vector Calculus & Differential Equations, Physiological Systems Modeling, Microprocessors & Assembly, Embedded Systems.
- Academic Focus:** Physiological Signal Dynamics and Medical Device Circuit Design.

KEY ENGINEERING PROJECTS

Microprocessor-Based Signal Processing Engine

Academic Research

- Simulated an 8-bit instruction cycle loop using MATLAB to process simulated biological sensor inputs.
- Implemented modular register files and ALU functions to perform real-time diagnostic calculations.
- Documented system architecture and execution routines for medical monitoring prototyping.

KEY ENGINEERING PROJECTS (CONTINUED)

Arduino Temperature & Environmental Monitoring System

Prototyping Project

- Prototyped a closed-loop temperature control fan circuit using Tinkercad to monitor environmental conditions for sensitive biological equipment.
- Integrated digital temperature sensors with feedback control loops to maintain fixed thermal thresholds.
- Tested circuit responsiveness to rapid thermal fluctuations, maintaining set parameters within $\pm 0.5^{\circ}\text{C}$.

Medical Data Visualization & Dashboard Design

Data Analytics Project

- Developed dynamic dashboards incorporating statistical modeling, Pareto distributions, and frequency histograms for biomedical datasets.
- Formatted key health indicators to streamline analytical reporting and clinical decision-making workflows.

TECHNICAL COMPETENCY MATRIX

Domain	Tools / Frameworks	Application Area
Signal Processing	MATLAB, Python	Filtering, FFT Analysis, Physiological Waveforms
Circuit Prototyping	Arduino, Tinkercad, Proteus	Biosensor Interfacing, Microcontrollers
Web & Interface	HTML5, CSS3, JavaScript	Portfolio & Clinical Dashboard Layouts
Quality & Compliance	Biomedical Standards, ISO 13485 concepts	Safety Assessment & Equipment Calibration

CERTIFICATIONS & AFFILIATIONS

- **Member:** Biomedical Engineering Society (Student Chapter)
- **Workshop Certification:** Circuit Prototyping & Medical Sensor Integration
- **Technical Seminar:** Advances in Medical Diagnostics & Clinical Engineering Practices

SOFT SKILLS & PROFESSIONAL ATTRIBUTES

- **Analytical Problem Solving:** Adept at breaking down complex biological systems into structured mathematical models.
- **Cross-Disciplinary Communication:** Capable of bridging technical terminology between engineers, healthcare staff, and researchers.
- **Technical Documentation:** Experienced in creating clear circuit schematics, system flowcharts, and research summaries.